

Web Services Security (WS-Security) via Oracle Web Services Manager (OWSM)

An architectural overview, including context with
Oracle Access Manager (OAM)

Agenda

- HTTP Security vs. WS-Security
- Components of WS-Security
- WS-Security Standards
- Benefits of Centralized WS-Security
- Benefits of OWSM Specifically
- Where OAM Fits In
- OWSM with OAM - Architecture
- Don't drive your motorcycle naked!!



HTTPS Security vs. WS-Security

- HTTPS is Transport-level Security (facilitated by OAM)
 - NOT End-To-End Security
 - Message is ONLY secured “On the Wire”
 - Message is not created securely
 - Message is not consumed securely
- Web Services Security is Message-level Security (facilitated by OWSM)
 - End-To-End Security
 - Message creation is secured
 - Message is consumed and distributed securely
 - Can work with “On the Wire” security, but DOES NOT RELY on it



Components of WS-Security

- Authentication
 - Without any proper identification of the user, it will be impossible to perform an audit of any unauthorized access
- Authorization
 - Without proper privileges for the user, it will be impossible to perform the operation.
- Confidentiality
 - Without proper XML message encryption, sensitivity of parts of messages cannot be protected
- Integrity
 - Without digitally signing the messages, they cannot be made tamper proof

Note: OAM does each extent, but not nearly to the robustness of OWSM



WS-Security Standards

- XML Encryption and XML Signature forms the basis for WS-Security
- Follows W3C Standards
- Includes Security Token Profiles
 - Username
 - SAML Tokens
 - Kerberos
 - X509

Benefits of Centralized WS-Security

- Eliminates complexity for development
 - No need to code encryption/decryption, digitally signing, etc.
- Allows central administration/management of security to multiple services
- Allows definition and reuse of Security Policies
 - Logical grouping of security characteristics specific to subsets of web services based on security/risk exposure
 - Extremely robust security apparatus modularized and separated away from applications/services
- Possibility to move security processing to another server

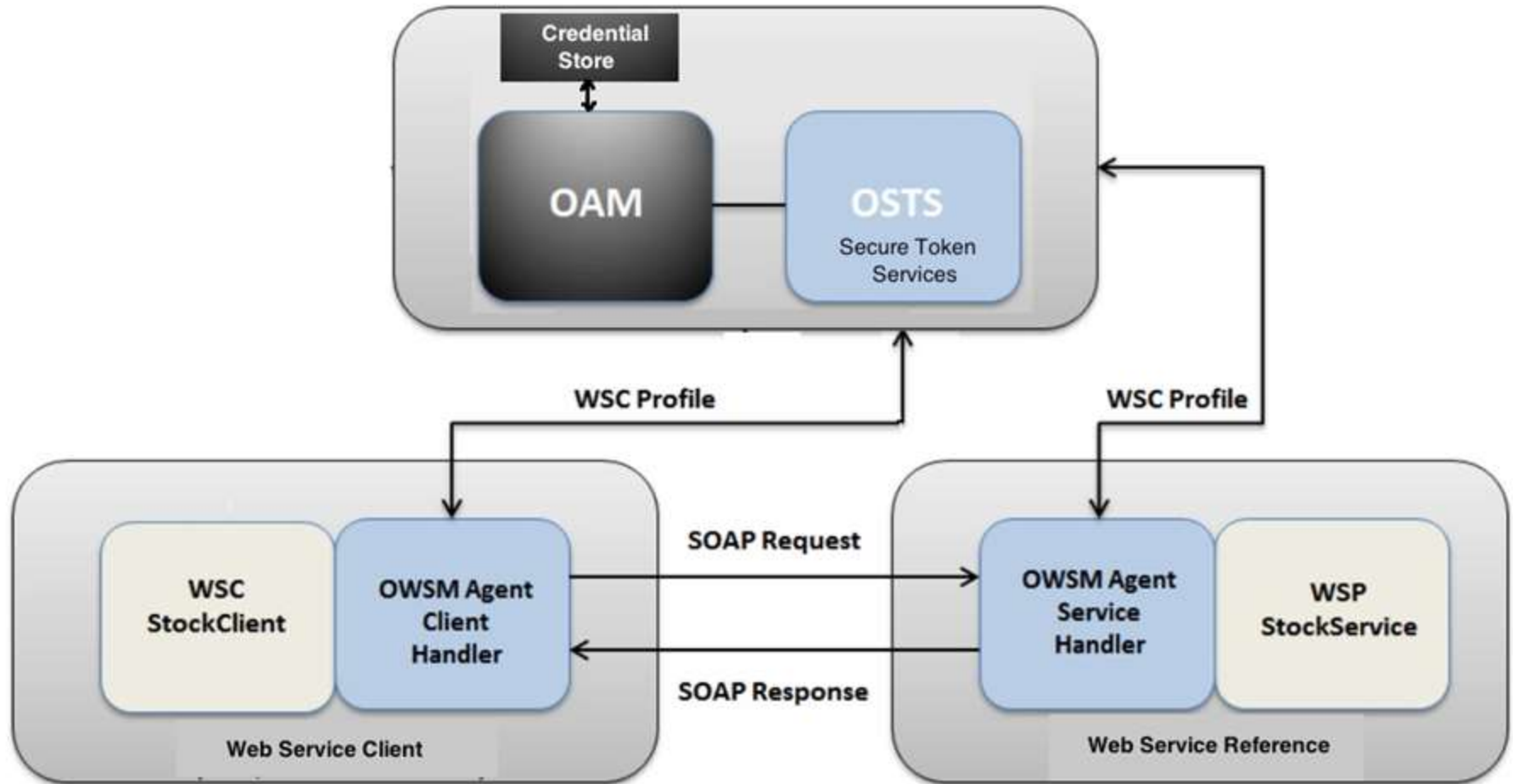
Benefits of OWSM Specifically

- All the benefits of generic WS-Security and more
 - Authentication
 - Centralizing security operations
 - Defining different policies for different web services
 - Creating policy templates that can be applied over a set of services
 - Enhanced logging capability for troubleshooting
 - Flexible auditing mechanism to track the performance
 - Strong integration with Oracle Access Manager (OAM) and CA SiteMinder
 - Flexible architecture for easy customization
 - Monitor web service operations, security operations, etc.
 - Manage the web services deployment
 - Generate reports about web services performance, usage, etc.

Where OAM Fits In

- OAM basic functions
 - Access management
 - Identity Federation
 - Secure Token Services
- OWSM integrates tightly with the following OAM functions
 - Secure Token Services
 - Secure Token Acquisition
 - Secure Token Validation
 - Secure Token Renewal

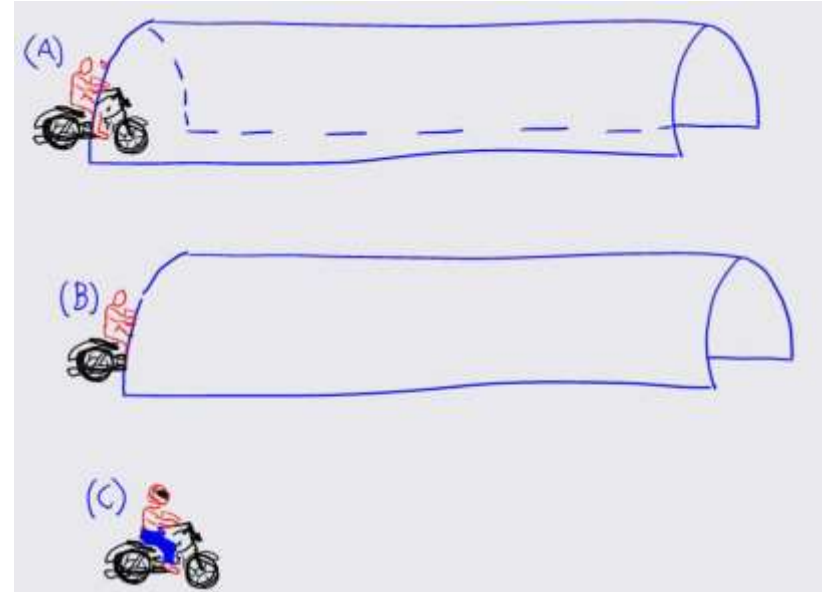
OWSM with OAM - Architecture



OWSM Agent intercepts requests and responses and executes Policies after mapping them

Don't drive your motorcycle naked!!

- (A): Naked Motorcyclist in a Transparent Tunnel (HTTP)
- (B): Naked Motorcyclist in a Opaque Tunnel (HTTPS)
- (C): Clothed Motorcyclist (WS-S)



Being clothed, the motorcyclist can leave home and travel all the way to his destination – including taking detours – without risking being exposed along the way. He need not rely upon any opaque tunnel to cover up!

(Image from Cloud Identity)